

Bias and Representation

Project: Verified Study Buddy

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Copyright and Provenance

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Red Team the Product

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prototype a Grounded Assistant

Without opening the old lesson, explain one key idea from Prototype a Grounded Assistant. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Representation and bias evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Representation and bias evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

An illustration set always shows scientists in the same role and appearance. Define a review that examines representation without inferring personal identities from images.

6. Solve the given case

Scientist images repeatedly show only one demographic group.

Evaluate and improve representation.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

An assistant is tested on six fluent requests but no requests with spelling errors. What gap does this leave, and how should the evaluation expand?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Audit answer quality across reading levels, languages, names, and accessibility needs.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

An illustration set always shows scientists in the same role and appearance. Define a review that examines representation without inferring personal identities from images.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

13. W2 Trace the worked case

A grounded assistant supports 9 of 10 group-A cases and 6 of 10 group-B cases. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

14. W3 Audit the limit

A report omits this qualification: Group comparisons require task relevance and appropriate data governance; differences alone do not identify a causal mechanism. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

15. W4 Apply a new case independently

Group C has 8 supported answers out of 10; D has 9 out of 15. Compute rates and difference; explain why raw correct totals mislead.

My reasoning and evidence

16. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Bias and Representation

Compare study-answer support rates when groups have different numbers of questions.

Group Birch has 6 supported answers in 8 cases; group Pine has 9 supported answers in 12 cases. Both groups use the same support rubric.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Bias and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Keep denominators

Birch: $\frac{6}{8}$. Pine: $\frac{9}{12}$.



2. Compare rates

Both support rates are 75%; Pine has more correct answers because it has more cases.



3. Inspect errors

Check the two Birch and three Pine failures before deciding what revision each needs.

CHANGE ONE THING

Pine instead has 6 supported answers out of 12.

Explain the changed result using the matching lesson.

Explain why comparing six correct answers in each group would mislead. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

19. Project portfolio milestone

Project: Verified Study Buddy

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare study-answer support rates when groups have different numbers of questions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Check the two Birch and three Pine failures before deciding what revision each needs.

1. Compare rates
2. Inspect errors
3. Keep denominators

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect errors Check the two Birch and three Pine failures before deciding what revision each needs. Read the picture in order. Check the two Birch and three Pine failures before deciding what revision each needs. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed-case practice: Pine instead has 6 supported answers out of 12. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Pine is 50%, which is 25 percentage points below Birch.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Pine is 50%, which is 25 percentage points below Birch. Equal correct counts do not imply equal rates when denominators differ.

Repair a missing building block

Review Copyright and Provenance (ai-3/29). Goal: Explain the mechanism and terminology of copyright and permitted sources. Short practice: Does acknowledging Fern make a cropped version permitted? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the mechanism and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Prototype a Grounded Assistant (ai-3/24). Goal: Explain the mechanism and terminology of capstone grounded assistant. Short practice: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

